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Additive Manufacturing Modification by Artificial Intelligence, Machine Learning, and Deep Learning: A Review

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Abstract: The manufacturing sector has been transformed owing to additive manufacturing (AM), which has made it possible to create intricate, personalized items with little material waste. However, optimizing and enhancing AM processes remain challenging owing to the intricacies involved in design, material selection, and process parameters. This review explores the integration of artificial intelligence (AI), machine learning (ML), and deep learning (DL) techniques to improve and innovate in the field of AM. AI-driven design optimization procedures offer innovative solutions for the 3D printing of complex geometries and lightweight structures. By leveraging machine learning (ML) algorithms, these procedures analyze extensive data from previous manufacturing processes to enhance efficiency and productivity. ML models facilitate design and production automation by learning from historical data and identifying intricate patterns that human operators may miss. Deep learning (DL) further augments this capacity by utilizing sophisticated neural networks to manage and interpret complex information and provide deeper insights into the manufacturing process. Integrating AI, ML, and DL into AM enables the creation of optimized, lightweight components that are crucial for reducing fuel consumption in the automotive and aviation industries. These advanced AI techniques optimize the design and production processes and enhance predictive modeling for process optimization and defect detection, leading to improved performance and reduced manufacturing costs. Therefore, integrating AI, ML, and DL into AM improves precision in component fabrication, enabling advanced material design innovations and opening new possibilities for innovation in product design and material science. This review discusses and highlights significant advancements and identifies future directions for applying AI, ML, and DL in AM. By leveraging these technologies, AM processes can achieve unprecedented levels of precision, customization, and productivity for analysis and modification.

Keywords: Additive manufacturing; Optimization; Artificial intelligence; Machine learning; Deep learning

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1. Introduction

3D printing, or additive manufacturing (AM), is a revolutionary industrial production method that enables the production of stronger and lighter systems and parts. AM technologies are classified according to a standardized framework defined by ISO/ASTM 52900. This framework organizes AM processes based on material models that are added or manipulated during the building process [1]. The widely accepted classification comprises the following seven categories:

- Material Extrusion: A material is selectively dispensed through a nozzle or orifice, as in fused deposition modeling (FDM).
- Binder Jetting: A liquid bonding agent is selectively deposited to join powder materials, such as binder jetting for metal or sand.
- Powder Bed Fusion: Thermal energy selectively fuses regions of a powder bed, such as in selective laser sintering (SLS) and direct metal laser sintering (DMLS).
- Vat photopolymerization: A liquid photopolymer in a vat is selectively cured by light-activated polymerization such as stereolithography (SLA).
- Material Jetting: Material droplets are selectively deposited via PolyJet and MultiJet.
- Sheet Lamination: Sheets of material are bonded to form an object (e.g., in laminated object manufacturing (LOM)).
- Directed Energy Deposition (DED): Focused thermal energy is used to fuse materials by melting while they are being deposited (e.g., in laser engineering net shaping, LENS).
- This standardized classification provides a clearer understanding of the technology, categorizing it based on the core processes used rather than specific brand names or techniques such as SLA or FDM.

Artificial intelligence (AI) is an overarching field encompassing the development of machines and systems capable of performing tasks that typically require human intelligence, such as decision-making, problem solving, and language understanding [2]. Machine learning (ML) is a specific subset within AI that enables systems to learn from data, identify patterns, and make decisions with minimal human intervention [3]. Deep learning (DL), a further specialization within ML, involves the use of neural networks with multiple layers (deep neural networks) to model complex data representations and improve accuracy in tasks such as image recognition and natural language processing [4]. ML focuses on algorithms that learn from data to make predictions or decisions, whereas DL uses more advanced neural network architectures capable of processing complex information and deriving deeper insights [3]. DL techniques, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have been employed to handle intricate patterns and enhance predictive models for process optimization and defect detection [5]. Although ML and DL are becoming integral parts of AI, each contributing to the enhancement of the capabilities of AI systems, AI still spans a broad spectrum of technologies and approaches [6]. AM has revolutionized traditional manufacturing processes by enabling rapid prototyping, customized production, and intricate design capabilities. However, advancements in AI, ML, and DL are driving AM to reach its full potential. AM capabilities can be significantly enhanced when combined with AI, ML, and DL. These technologies are reshaping the AM landscape by enhancing process efficiency, material property optimization, and design complexity management [7]. These technologies provide opportunities for smarter and more efficient manufacturing systems by facilitating the discovery of novel materials, increasing quality control, streamlining production, and improving design processes [8]. These methods have shown great promise for a

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variety of applications, including image processing, natural language processing, and autonomous systems. AM operations can benefit from the application of AI, ML, and DL to improve quality control, predict material behavior, and optimize process parameters [9].

AI is fundamentally altering the functioning and development of AM systems. Large-scale datasets produced during 3D printing can be analyzed using AI algorithms, which makes it possible to optimize important factors, such as print speed, material consumption, and structural integrity. This integration has led to enhanced precision and consistency in manufacturing outcomes [10]. ML plays a critical role in forecasting AM equipment maintenance requirements using past and current data based on usage trends and operational indicators. Predictive maintenance reduces downtime and increases output [11]. Furthermore, ML algorithms can enhance supply chain efficiency by reducing overproduction and averting stockouts in terms of anticipating material and work demands [12]. ML optimizes routing and scheduling in logistics, ensuring timely deliveries, and reducing transportation costs [13]. Furthermore, ML enhances inventory management by accurately predicting consumption patterns and lead times, thereby maintaining optimal inventory levels [14]. DL, a subset of ML, excels in handling complex pattern recognition and data modeling tasks, which is particularly effective for defect detection and process monitoring through advanced image and pattern recognition. DL models can identify subtle anomalies during the printing process, thereby enabling immediate corrective actions to prevent defects [15]. Moreover, DL enables predictive maintenance by forecasting equipment failures based on intricate correlations within the operational data, thereby facilitating proactive maintenance scheduling [16]. By integrating AI with AM systems, manufacturers can achieve real-time monitoring and management of printing operations [2, 17]. AI-driven systems can make swift adjustments to process parameters, ensuring ideal printing conditions and significantly reducing the likelihood of errors [7, 18]. Moreover, AI automates design and manufacturing tasks, streamlines process optimization, and anticipates potential failures, thereby enhancing the overall efficiency and reliability of 3D-printing operations [19]. Fig. 1 shows the application of data-driven ML in 3D printing [20]. 3D printing designing rules are presented in Fig. 2 [21].

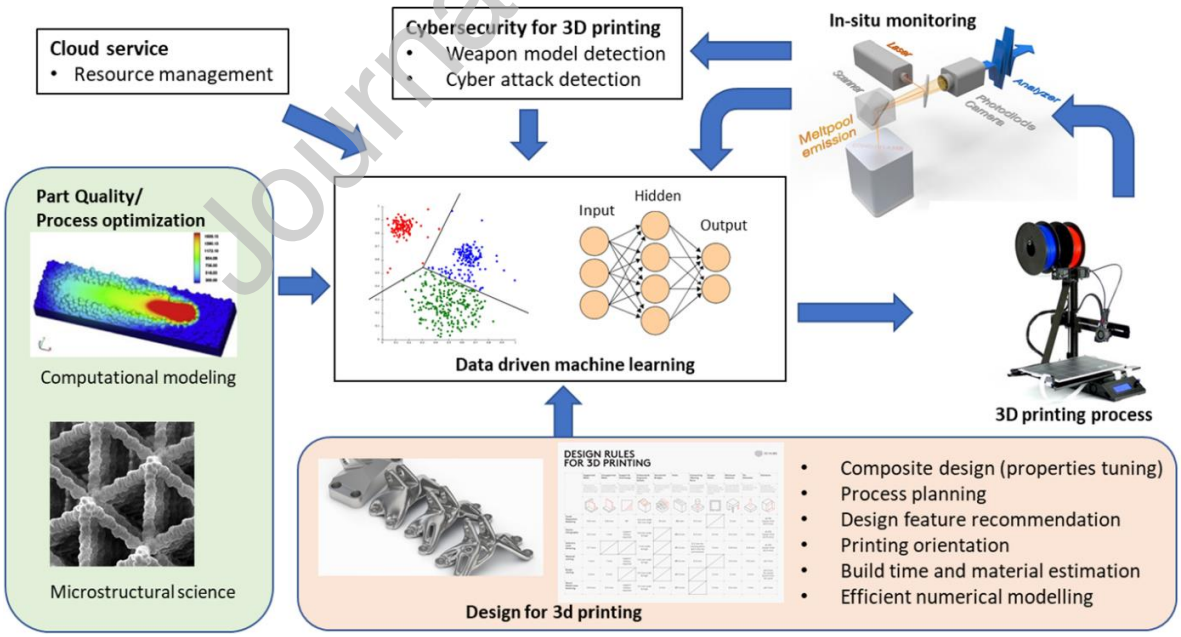

Fig. 1. Application of data driven ML in the 3D printing process [20].

	Supported walls	Unsupported walls	Support & overhangs	Embossed & engraved details	Horizontal bridges	Holes	Connecting /moving parts	Escape holes	Minimum features	Pin diameter	Tolerance
	Walls that are connected to the rest of the print on at least two sides.	Unsupported walls are connected to the rest of the print on less than two sides.	The maximum angle a wall can be printed at without requiring support.	Features on the model that are raised or recessed below the model surface.	The span a technology can print without the need for support.	The minimum diameter a technology can successfully print a hole.	The recommended clearance between two moving or connecting parts.	The minimum diameter of escape holes to allow for the removal of build material.	The recommended minimum size of a feature to ensure it will not fail to print.	The minimum diameter a pin can be printed at.	The expected tolerance (dimensional accuracy) of a specific technology.
Fused deposition modeling	0.8 mm	0.8 mm	45°	0.6 mm wide & 2 mm high	10 mm	Ø2 mm	0.5 mm		2 mm	3 mm	±0.5% (lower limit ±0.5 mm)
Stereo-lithography	0.5 mm	1 mm	support always required	0.4 mm wide & high		Ø0.5 mm	0.5 mm	4 mm	0.2 mm	0.5 mm	±0.5% (lower limit ±0.15 mm)
Selective laser sintering	0.7 mm			1 mm wide & high		Ø1.5 mm	0.3 mm for moving parts & 0.1 mm for connections	5 mm	0.8 mm	0.8 mm	±0.3% (lower limit ±0.3 mm)
Material jetting	1 mm	1 mm	support always required	0.5 mm wide & high		Ø0.5 mm	0.2 mm		0.5 mm	0.5 mm	±0.1 mm
Binder jetting	2 mm	3 mm		0.5 mm wide & high		Ø1.5 mm		5 mm	2 mm	2 mm	±0.2 mm for metal & ±0.3 mm for sand
Direct metal Laser sintering	0.4 mm	0.5 mm	support always required	0.1 mm wide & high	2 mm	Ø1.5 mm		5 mm	0.6 mm	1 mm	±0.1 mm

Fig. 2. Designing rules for 3D printing [21].

Algorithms based on AI and ML can optimize manufacturing and design parameters, improving material use, reducing production times, and improving product quality [22]. For example, AI-driven generative design can examine a wide range of design options and identify the best combinations that satisfy certain performance requirements. Additionally, ML models can anticipate any flaws and recommend remedial actions during the production process, guaranteeing improved uniformity and dependability [23, 24]. DL, which is a subset of ML, offers powerful capabilities for processing and interpreting complex data, such as images and sensor readings. DL algorithms can be utilized in AM for real-time monitoring and control, enabling in situ changes in process parameters and anomaly detection [25]. Implementing a real-time feedback loop could substantially decrease the frequency of faults and improve the overall quality of manufactured goods [26]. By analyzing large datasets, finding patterns, and making accurate predictions, these cutting-edge computational approaches of AI, ML, and DL can improve the effectiveness, dependability, and general performance of AM operations [27].

Karimi Ghaleh Jough and Şensoy [28] introduced meta-heuristic methods to estimate the bursting risk of mid-rise steel tension frame buildings, thereby improving risk management in concrete moment frames. Karimi Ghaleh Jough and Golhashem [29] investigated the out-of-plane motion of ad hoc brick structures using finite element models and found that walls built with the lightest masonry components currently available in the market had less self-weight axial distortion. Karimi Ghaleh Jough and Beheshti Aval [30] developed an adaptive neuro-fuzzy inference framework using the fuzzy C-means method to construct the seismic sensitivity curve for an SMRF structure. Consequently, the computational accuracy increased, and epistemic uncertainty was incorporated. Unlike the previously described methods and the unpredictable nature of epistemic knowledge in employing a group-based data processing approach, Karimi Ghaleh Jough et al. [31] exploited vulnerabilities to increase power and output precision while maintaining the same processing time. Karimi Ghaleh Jough and Ghasemzadeh [32] provided

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the foundation for constructing a 3D-fragility curvature to assess the unknown interval of steel at that instant. The study aimed to use optimal fuzzy approaches designed to decrease the execution time and increase precision while establishing a 3D-fragility curve. Karimi Ghaleh Jough [33] examined the impact of steel wall posts on the out-of-plane behavior of nonstructural concrete walls to create wall posts for masonry walls with fewer adjustment parts. Soori et al. [34] studied sustainable CNC machining operations to reduce environmental contamination during the manufacturing process. Soori and Karimi Ghaleh Jough [35] assessed AI application to steel moment-frame construction optimization to enhance the operational performance of these structures.

Soori et al. [36-40] recommended virtual machining methods for improving and assessing CNC machining in virtual settings. Soori et al. [41] recommended using a complex virtual machining method to improve the surface characteristics when milling turbine blades in five-axis milling operations. Soori [42] employed virtual innovation to better understand and investigate the part creation process in virtual environments. Soori and Karimi Ghaleh Jough [43] obtained optimized machining parameters to assess and eliminate the surface roughness and residual stress during EDM machining processes. Dastres et al. [44] reviewed RFID-based wireless manufacturing systems to improve component manufacturing accuracy and dependability, data availability and quality across the supply chain, and energy consumption efficiency. Soori et al. [44] investigated the potential advantages of AI and ML in CNC machine tools for boosting productivity and profitability in the element production industry. Soori and Arezoo [45] examined the problem of determining and minimizing the residual stress in machining procedures to enhance the function of machined parts.

Dastres and Soori [46] recommended utilizing communication channels for environmental concerns to reduce the negative effects of technological advancements on calamities. Similarly, Dastres and Soori [47] suggested using the secure socket layer to strengthen network and data security on the internet.

Dastres and Soori [48] examined the developments in web-based decision support systems and proposed the gaps between various approaches to develop a decision support system methodology. Dastres and Soori [50] reviewed the most recent advancements in network threats to improve network security techniques. Soori et al. [49] explored recent advances in the literature to assess and improve the impact of AI, DL, and ML on sophisticated robots.

Soori and Arezoo [50] investigated the effect of coolants on cutting temperature, tool use, and surface roughness when turning a Ti6Al4V material. Soori [51] conducted a survey of the current developments from published works to study and alter composite materials and structures. Soori et al. [52] investigated strategies to improve industrial quality control and streamline parts manufacture in Industry 4.0 smart factories with the internet of things (IoT) applications. Soori and Arezoo [55] studied virtual machining to reduce the amount of harm caused to drilling tools. Soori and Arezoo [53] reduced the surface roughness and residual stress of a product produced by water jet cutting to enhance its overall quality using an abrasive. Soori [54] detected and corrected possible deformation errors in turbine blade five-axis machining operations to improve precision. Soori et al. [55] reviewed blockchain applications in industrial IoT to increase the performance of sustainable supply chain management in Industry 4.0.

Soori et al. [56] conducted an investigation to assess and improve the Industry 4.0 component production process by examining the benefits and drawbacks of virtual manufacturing methodology. Soori et al. [57] studied AI use in developing supply chain management in high-tech manufacturing. Raoofi and Yildiz [58] examined battery state estimation techniques using machine learning for airplane propulsion battery management systems to enhance the battery management of the next

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generation of electric aircraft. Raoofi and Yasar [59] examined cutting-edge digital technologies in ongoing airworthiness management frameworks and apps to provide insights into the existing state of the interaction between maintenance procedures and the digital world and to draw attention to the untapped potential for digital change in aviation maintenance. Raoofi and Ölçen [60] highlighted the legal perspectives of continental aviation on environmentally friendly aircraft and airport infrastructure to guarantee the overall aviation sustainability.

Their previous publications provided an exhaustive framework for investigating the effects of AI and virtual simulations on advanced manufacturing procedures [61, 62]. Their research spans virtual machining, machining parameter optimization, tool wear prediction, and the integration of AI, DL, and ML into CNC machining and Industry 4.0 applications [63, 64]. The authors have applied AI to enhance manufacturing processes, such as surface characteristics improvement, tool life optimization, residual stress reduction, and machining accuracy refinement [65, 66]. These studies on AI-driven manufacturing systems, including digital twins, IoT-enabled smart factories, and blockchain applications for supply chain management, form a relevant backdrop for investigating the potential of AI in smart manufacturing processes [56, 67, 68]. By leveraging AI to streamline decision-making, increase efficiency, and improve process performance, the authors aim to apply these innovations to AM, enhancing the production quality, accuracy, and scalability of AM technologies within Industry 4.0 frameworks [55, 69].

This study aims to provide a comprehensive overview of DL, ML, and AI integration into AM. It examines the advancements in this interdisciplinary field, discusses the challenges encountered, and recommends additional research topics. This review seeks to provide a thorough overview of the current state of research on the use of ML, DL, and AI in AM. It examines the various applications and advantages of these technologies in AM, focuses on their limitations and restrictions, and discusses potential directions for further study and advancement. By analyzing the relationships between AM and these advanced methods (AI, ML, and DL), this study intends to demonstrate the possibilities for developing more intelligent, flexible, and efficient manufacturing systems. This study aims to illustrate the applications of new technologies, such as AI, ML, and DL, into AM through a review of several studies and case studies to suggest new ideas and future research directions. Table 1 categorizes the relationships and applications of the different methodologies in AM processes, design, and integration.

Table 1. The relationships and applications of different methodologies in AM processes, design, and integration.

Category	Subcategory	Description
1. Design optimization	Optimization of design	Improving design to achieve better performance and efficiency.
	Material development	Developing new materials to enhance product properties.
	Scalability and flexibility	Ensuring that design solutions can scale and adapt to various production levels and applications.
2. Process optimization	Processes optimization	Refining and improving processes for enhanced efficiency and reduced costs.

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	Process monitoring and control	Continuously tracking and adjusting processes to maintain optimal performance.
	Workflow Automation	Automating repetitive tasks to improve productivity and accuracy.
3. Quality management	Quality control and defect detection	Implementing measures to ensure the final product meets required standards and identifying defects.
4. Maintenance	Predictive maintenance	Using data to anticipate equipment failures and maintenance needs, preventing downtime.
5. Innovation & scalability	Innovation acceleration	Driving innovation by streamlining design, processes, and technologies.
	Adaptive learning	Using AI/ML systems to learn and optimize operations dynamically based on past data.
6. Supply chain optimization	Supply chain and workflow optimization	Streamlining supply chain and workflows for improved efficiency and reduced costs.
7. Sustainability	Sustainability in AM	Ensuring that processes, especially in additive manufacturing, are environmentally sustainable.

2. Design Optimization

AI-driven optimization techniques such as ML and DL have been increasingly utilized to enhance design process optimization during AM [7, 70]. These methods enable the generation of optimized geometries and material distributions that satisfy specific performance criteria and often surpass traditional design approaches [71]. The goal of this multidisciplinary strategy is to improve the capabilities, quality, and efficiency of the AM processes. In addition, AI systems can produce ideal design solutions, given a set of parameters and specifications [72]. AI can assist in generative design, in which algorithms explore all possible permutations of a solution and suggest the best options. Consequently, creative designs are frequently stronger and lighter than those created using conventional techniques [73]. AI algorithms can generate complex geometries that can be optimized for strength, weight, and material usage [74]. These designs often outperform human-created designs in terms of efficiency and functionality [75]. AI provides robust frameworks for improving design processes by leveraging algorithms that can simulate and predict outcomes more efficiently than traditional methods. AI techniques such as genetic algorithms and simulated annealing have been widely applied to optimize component design for 3D printing [76]. These techniques can handle multi-objective optimization problems, where the goal is to find the best trade-offs between conflicting objectives, such as strength, weight, and material usage. For example, AI can be used to optimize lattice structures in lightweight components, ensuring maximum strength while minimizing material usage [77]. AI systems can then identify the most efficient structures, which can be too computationally expensive to be manually obtained by iterating through several design configurations. [78]. Conversely, topology optimization focuses on optimizing the distribution of material within a given design space [79]. Consequently, topology optimization can achieve more efficient and innovative designs by incorporating ML and DL techniques [80].

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ML-powered topology optimization can produce strong, lightweight structures that are useful in the aerospace and automobile sectors. For example, General Electric used AI to improve fuel nozzle design, leading to weight reduction and better working life of nozzles [81]. Additionally, companies such as Airbus have applied AI-based generative design techniques to develop aircraft components, significantly reducing material usage while maintaining structural integrity [82]. These success stories demonstrate the tangible benefits of integrating AI into the design optimization phase of AM, leading to more efficient, cost-effective, and innovative products [83]. In addition, DL models can be used to forecast ideal patterns of material distribution, which improve mechanical qualities and require less material. ML can also optimize the topology of planned items by refining their geometry to use less material while preserving their strength and performance [84]. The topology optimization of the designed parts can be implemented using ML to reduce weight and material usage while maintaining strength. AI techniques can also optimize material distribution within a given design space, improving performance while reducing material usage. Design performance in different situations can be predicted using ML and DL models, which enable virtual testing and optimization before actual manufacturing [85]. ML algorithms can offer extensive skills for simulating complex connections between design inputs and manufacturing outcomes. Supervised learning techniques, such as regression analysis and support vector machines, can forecast the performance of a design based on historical data [86]. These models can then be used to identify optimal design parameters that meet specific performance criteria. In addition, reinforcement learning has shown promise for optimizing design parameters by treating the design process as a sequential decision-making problem [87]. Reinforcement learning algorithms can progressively improve design and achieve optimal solutions more efficiently by continuously learning from the results of previous design iterations. The ML-integrated design for AM is shown in Figure 3 [88]. An integrated ML framework can be used to determine the links between the process, structure, and property, as shown in Fig. 3(a) [88]. The ML model can be trained using the outputs of corresponding structural parameters, such as surface roughness, dimensional accuracy, and voids, by considering the process parameters from AM processes, such as print speed, print temperature, and layer thickness of the part [89]. The process-structure link is then captured by the trained ML model, which can be used in subsequent design phases. An ML-trained model can use structural parameters such as surface roughness and dimensional accuracy as inputs and AM process parameters such as print speed, print temperature, and layer thickness as outputs to determine the relationship between the structure and process [88]. Consequently, subsequent designs, from structural to procedural, can take advantage of the learned ML models. As shown in Fig. 3(b), further links between the process, structure, and property can be constructed using ML in either direction [88].

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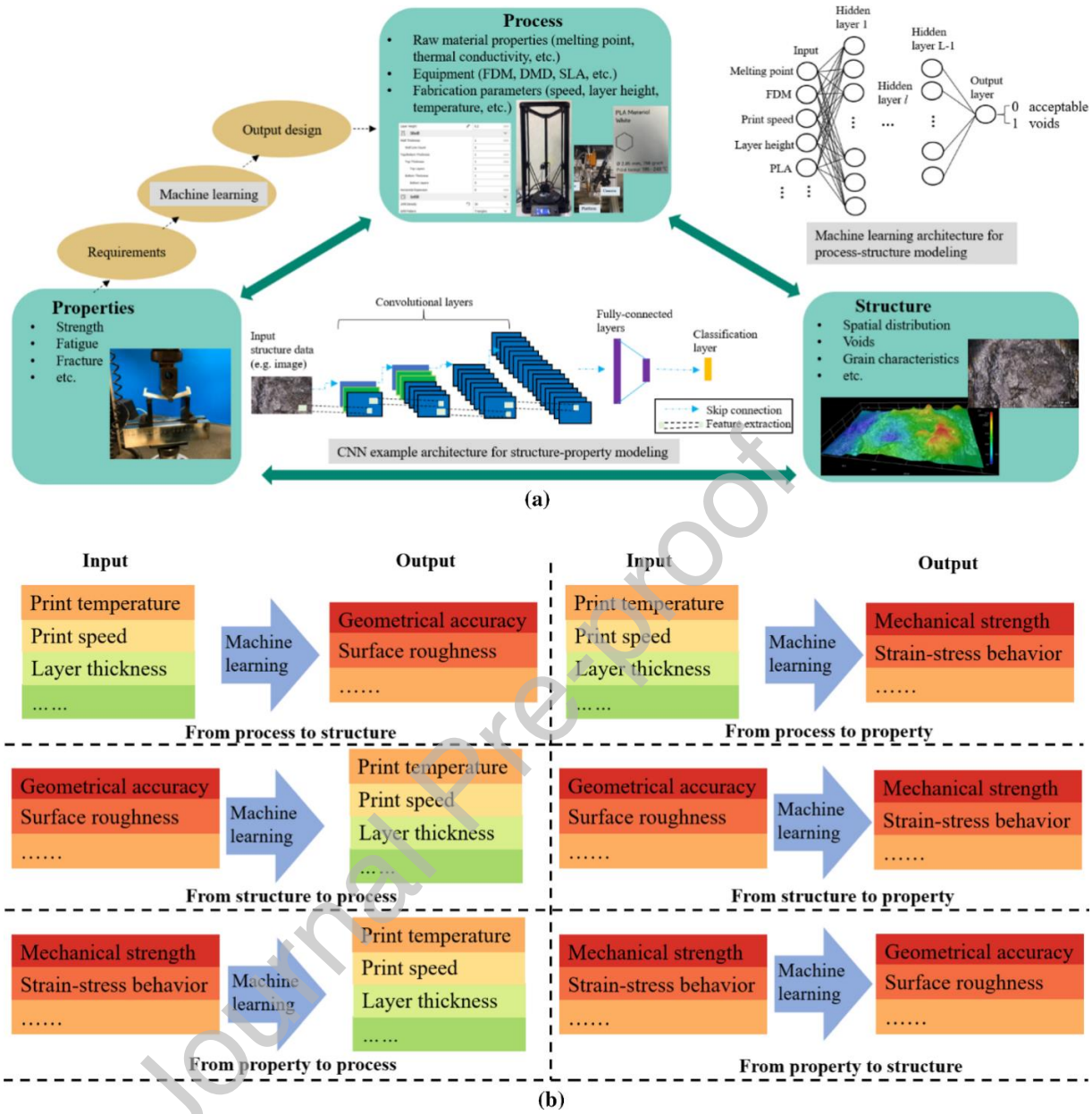

Fig. 3. ML-integrated design for AM [88].

DL, a subfield of machine learning, is particularly well-suited for AM design optimization owing to its ability to handle complex patterns and high-dimensional data. DL models can be applied to optimize the geometry of certain performance parameters, leading to creative designs. By integrating AI into the AM process, generative design generates several design iterations by considering structural integrity, weight, and material consumption [90]. ML algorithms can then evaluate these designs to identify the best solution [91]. For instance, Airbus utilized a generative design to create lightweight and structurally efficient components for its aircraft, resulting in significant weight savings and improved fuel efficiency [92]. Generative design algorithms can explore a vast design space, producing structures that are both lightweight and strong. Convolutional neural networks (CNNs) and generative adversarial networks (GANs) are two DL techniques that can be successfully applied in this

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context. CNNs are useful for both the structural analysis and topology optimization of components, ensuring that designs can be produced using AM technology and providing advanced performance in working conditions. Conversely, GANs can generate novel design variations by learning from a dataset of existing designs, thereby expanding the design space and potentially discovering innovative solutions that human designers may overlook. An overview of the research status and applications of AI and ML in AM processes across various areas such as design optimization, process optimization, quality control, defect detection, and predictive maintenance is presented in Table 2.

Table 2. The status and applications of AI and ML in AM process across various areas like design optimization, process optimization, quality control, defect detection, and predictive maintenance.

Research area	AI techniques/methods	Current research status	Applications
Design optimization	ML, Genetic Algorithms, DL	Widely researched; AI-driven design automation is being implemented in generative design	Improves part design efficiency, reduces time for complex geometries [93]
Processes optimization	Neural networks, Reinforcement learning, Bayesian optimization	Research focusing on optimizing printing parameters (e.g., speed, temperature)	Enhances production speed, material usage, and energy efficiency [94]
Quality control	Computer vision, CNNs, ML	Advanced studies on real-time monitoring of build quality	Ensures high-quality prints, minimizes post-processing requirements [95]
Defect detection	CNNs, Image recognition, ML, Data analytics	Active research on identifying and predicting defects during and after printing	Reduces defects in final products, increasing reliability [96]
Predictive maintenance	Predictive analytics, ML, Big data	Ongoing development to predict failures in machines and systems	Reduces downtime, improves machine lifespan, and cuts maintenance costs [97]
Material optimization	Data-driven models, AI algorithms, ML	Growing interest in optimizing material selection for AM	Helps identify material properties that optimize mechanical performance [98]

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Design optimization	ML, Genetic algorithms, DL	Widely researched; AI-driven design automation is being implemented in generative design	Improves part design efficiency, reduces time for complex geometries [99]

3. Processes Optimization

AI and ML algorithms can optimize AM processes by predicting the optimal set of process parameters that yield the desired material properties and minimize defects [100]. For instance, genetic algorithms and reinforcement learning can be used to explore the parameter space and identify optimal conditions. Applying AI, ML, and DL to optimize 3D printing processes involves employing these technologies to improve AM workflows at different phases [101, 102]. This includes design, material selection, process parameter optimization, real-time monitoring, and post-processing. AI can optimize AM process parameters such as temperature, speed, and material flow to improve product quality and reduce defects [103]. Further, AI and ML models can predict optimal printing parameters such as temperature, speed, and layer thickness for different materials and designs, reducing trial and error [76]. AI can also be used to analyze large datasets generated during AM processes to identify patterns and correlations that are not apparent to human operators. This can improve decision-making and process control [104]. ML algorithms can analyze data from various sources, including process parameters, sensor readings, and post-process inspections, to identify correlations between process variables and part quality [105]. Manufacturers can leverage this information to optimize process parameters, thereby minimizing defects and improving part quality [106]. ML models can predict the outcomes of AM processes, such as the mechanical properties of the final product, potential defects, and optimal process parameters [107]. ML models can analyze large datasets from previous print jobs to determine the optimal settings for new jobs, reducing trial and error and material waste [108]. DL can analyze the complex relationships between process variables and output quality to enhance overall efficiency. DL can simulate 3D printing processes to predict the outcomes of different parameter settings [103]. Therefore, this virtual experiment reduces the need for physical trials, saving time and resources. Furthermore, applying AI and ML to optimize AM processes can promote sustainability and energy efficiency. AI can reduce the environmental effects of 3D printing by anticipating and reducing energy use throughout the printing process. Furthermore, ML algorithms can optimize material usage, thereby reducing waste and promoting sustainable manufacturing practices [109]. Thus, AI, ML, and DL integration into AM still has considerable room for improvement. Manufacturers can increase the precision, efficiency, and dependability of the manufacturing process by utilizing these technologies [110].

4. Quality Control and Defect Detection

Quality assurance is a crucial aspect of AM that must be achieved during part production. Ensuring the quality of AM-produced components is crucial, especially for applications in the automotive, aerospace, and medical domains. [111]. These specific component properties are the direct outcome of integrating AI and ML into the AM process and connecting them with the designed AI algorithms [112]. Manufacturers can focus on and enhance certain component characteristics more efficiently by customizing algorithms for particular process inputs and outputs [113]. AI, ML, and DL algorithms have been applied to optimize the following specific properties for enhancement:

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- Dimensional accuracy: AI-based predictive models can dynamically adjust process parameters, such as laser power, scan speed, and layer height, to reduce deviations from the desired dimensions [114].
- Surface roughness: ML algorithms analyze process parameters and identify conditions that minimize surface roughness, thereby improving post-processing quality and reducing the need for additional surface treatments.
- Mechanical properties: DL models can predict optimal printing strategies that lead to enhanced mechanical performance, such as increased tensile strength, fracture toughness, and hardness, by adjusting parameters, such as cooling rates and deposition paths [115].
- Microstructural control: AI can be employed to predict and control grain structure and phase distribution during the building process, ensuring that the required material properties are achieved for specific applications, such as aerospace and medical devices [116].

Printing faults can be predicted using ML models trained on historical data. Furthermore, printed components can be inspected for flaws, such as porosity, surface roughness, and dimensional errors, using computer vision systems driven by DL algorithms [66, 117]. These automated inspection devices enhance the overall quality assurance process by reducing the need for manual inspection and reworking. Real-time data analysis from the AM process can also be implemented using ML algorithms to identify abnormalities that indicate flaws [118]. This facilitates prompt adjustments, reducing waste and enhancing product dependability. Using ML to monitor and control AM processes in real time enables component manufacturing with greater consistency and quality [119]. DL algorithms can analyze images from in situ monitoring systems to detect defects in real time, enabling immediate corrective actions. DL can also be used to analyze the 3D scan data of printed parts to ensure adherence to the design specifications for the produced components [120]. DL can improve defect detection accuracy by learning from a vast amount of image data and recognizing subtle patterns. Moreover, DL models can identify anomalies in sensor data, thereby ensuring consistent quality [121]. DL techniques, such as CNN, have been applied in real-time monitoring systems to enhance quality control by detecting anomalies in layer deposition, which could lead to defects in the produced parts [122]. GE Aviation using an AI-driven design for fuel nozzles is a notable successful case. ML models refined the design for AM, reducing the weight and enhancing the performance of the nozzles, which were successfully implemented in their LEAP engine [123]. AI can also adjust process parameters in real time based on sensor feedback to ensure optimal conditions [124]. Furthermore, AI can create closed-loop systems to adjust parameters based on real-time data to correct errors during the printing process [125]. In addition, AI can implement ML models to detect defects and anomalies in printed parts through image analysis and other nondestructive testing methods. Figure 4 shows a real-time defect detection strategy for AM processes based on DL and machine vision technologies [126].

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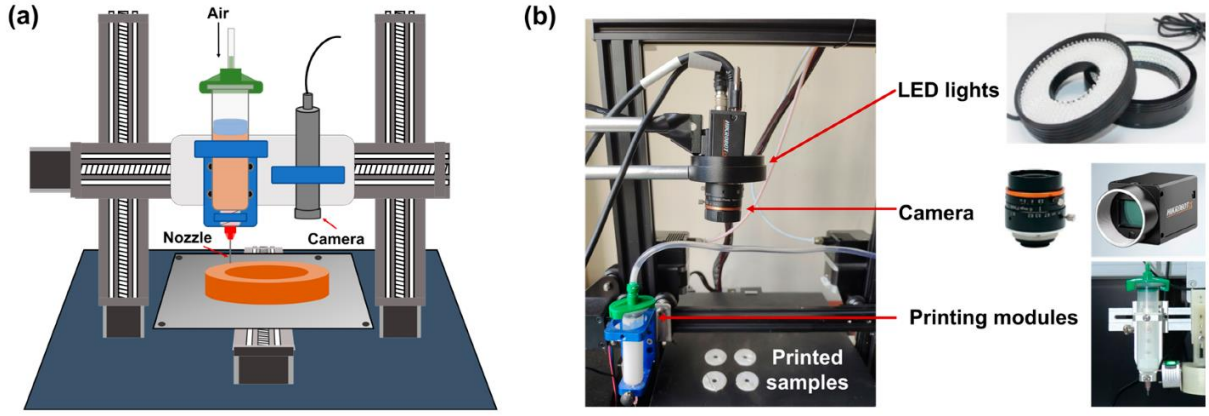

Fig. 4. A real-time defect detection strategy for AM processes based on DL and machine vision technologies [126].

Digital twin technology in metal printing revolutionizes the manufacturing process by creating a virtual replica of the physical printing environment [127]. This digital equivalent provides precise control over factors such as temperature, pressure, and material flow by enabling real-time monitoring and simulation of the metal printing process [128]. Digital twins enable the dynamic optimization and predictive modeling of AM processes by integrating AI, ML, and DL. AI-driven models can analyze large datasets generated by digital twins, facilitating continuous improvements in design, material selection, and process parameters, leading to improved efficiency and reduced production errors [129]. This synergy between digital twins and AI-based technologies advances AM precision and customization and fosters innovation by enabling adaptive manufacturing systems that learn and evolve over time. Digital twins can improve production parameters, predict potential issues, and improve the overall quality of printed metal components by applying physical system data [130]. The integration of digital twins enhances the efficiency and reliability of metal printing by reducing waste, lowering costs, and accelerating the development of innovative metal products. Figure 5 shows a digital twin in 3D printing [131].

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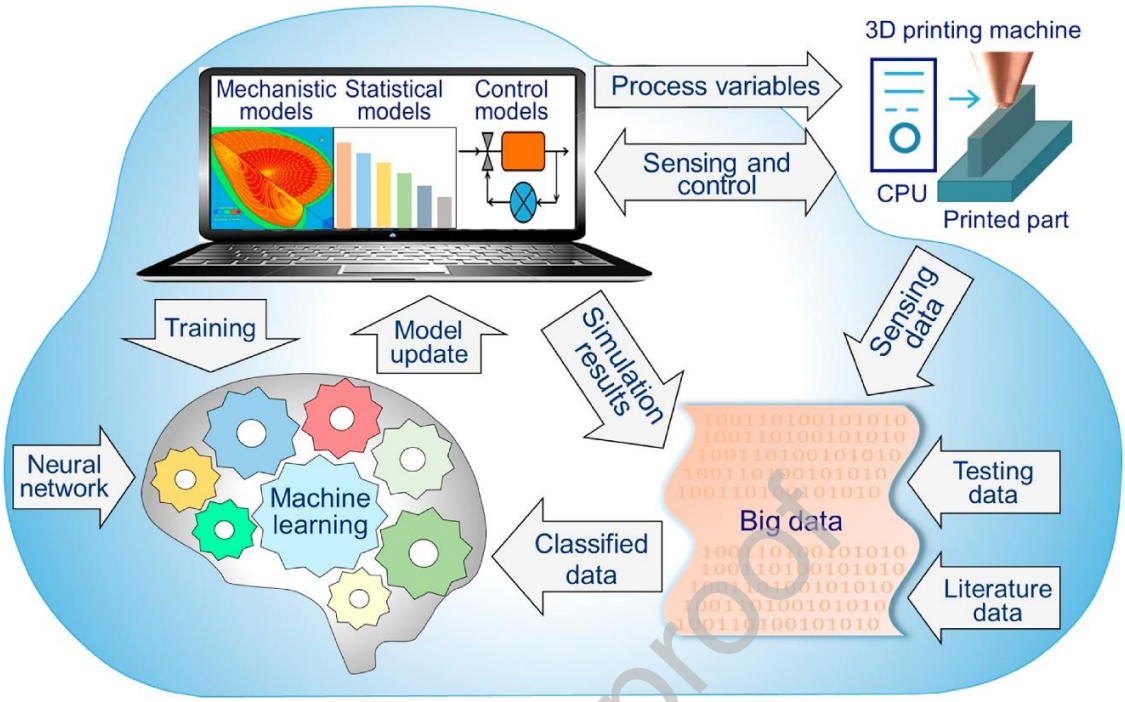

Fig. 5. Digital twin in 3D printing process [131].

Consequently, quality control can be improved using AI and DL algorithms for automated inspection and flaw identification during 3D printing.

5. Predictive Maintenance

Predictive maintenance is a key area of AM that uses AI, ML, and DL to increase the effectiveness and dependability of 3D printing [11]. ML can avoid downtime and prolong machine life by predicting when AM equipment requires repair [132]. AI algorithms can process vast amounts of data generated during the AM process, identifying patterns and anomalies that might indicate potential failures [19, 133]. AI systems can then make informed decisions regarding the timing of maintenance based on predictive analytics [134]. ML can also focus on developing algorithms that allow computers to learn from and make predictions based on data. Sensors and IoT devices collect data on machine performance, including temperature, vibration, and operational efficiency. ML models can thus analyze historical data to identify patterns that precede failures [135]. Therefore, these models can predict future failures based on the current machine conditions and historical data, thereby enabling proactive maintenance scheduling [136]. In addition, DL models can handle large volumes of complex data, such as images from machine vision systems and high-dimensional sensor data [137]. DL can then automatically extract relevant features from the raw data, which can be crucial for identifying subtle signs of wear or degradation. Therefore, DL models can detect anomalies in real time and provide immediate alerts for potential issues [138]. Consequently, predictive maintenance minimizes unexpected machine downtime by scheduling maintenance activities during non-productive periods [139]. These technologies help maintain the quality of manufactured parts by predicting and preventing machine failures, thereby reducing waste and rework [140]. By predicting and preventing machine failures, these technologies help maintain the quality of manufactured parts,

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thereby reducing waste and rework [141]. Predictive maintenance minimizes unexpected machine downtime by scheduling maintenance activities during non-productive periods [142].

6. Material Development

Innovative materials for AM can be developed more rapidly using AI and ML. These technologies can guide the development of novel materials with certain properties by analyzing the relationships between material composition, process parameters, and component performance [143]. Integrating AI, ML, and DL into AM processes can improve material development, boost production efficiency, and optimize the overall manufacturing workflow [144]. ML models can also predict the properties of new materials, accelerating the development of advanced materials with desirable characteristics [25, 145]. AI algorithms can predict the properties of new materials, reducing the need for trial and error. This accelerates the discovery of materials with desirable characteristics, such as strength, flexibility, or heat resistance [146]. Moreover, AI can design composite materials with tailored properties for enhanced performance in specific applications [101]. Furthermore, AI can optimize material usage in the AM process, thereby minimizing waste and cost. ML models can analyze large datasets of material compositions and their properties, identifying patterns and relationships that can guide the development of new materials [147]. A subset of ML, called reinforcement learning, can be very useful in identifying novel material compositions with improved performance applications [148]. Figure 6 illustrates layer-wise relevance propagation via a trained neural network [149].

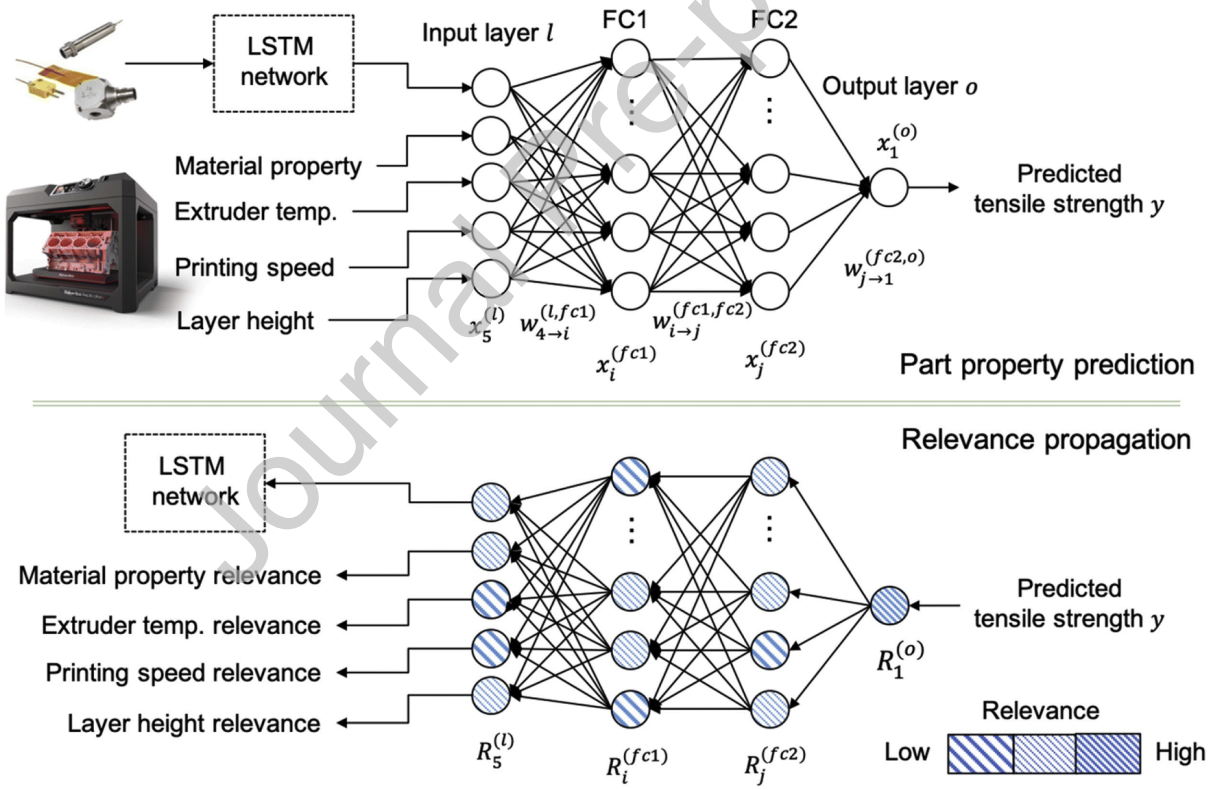

Fig. 6. Layer-wise relevance propagation via a trained neural network [149].

AI can also create adaptive manufacturing systems that can switch between different products and materials with minimal reconfiguration. Developing lightweight, high-strength materials tailored for specific components in the aerospace and

Journal Pre-proof

automotive industries can enhance the performance of parts produced using AM [101, 144]. Furthermore, customized implants and prostheses with material qualities designed for durability and biocompatibility are used in medical equipment [150]. Thus, enhanced design efficiency, reduced material usage, and improved product performance are some advantages of applying AI, ML, and DL in AM processes [151, 152]. Figure 7 shows AI-assisted material development for the AM process [153].

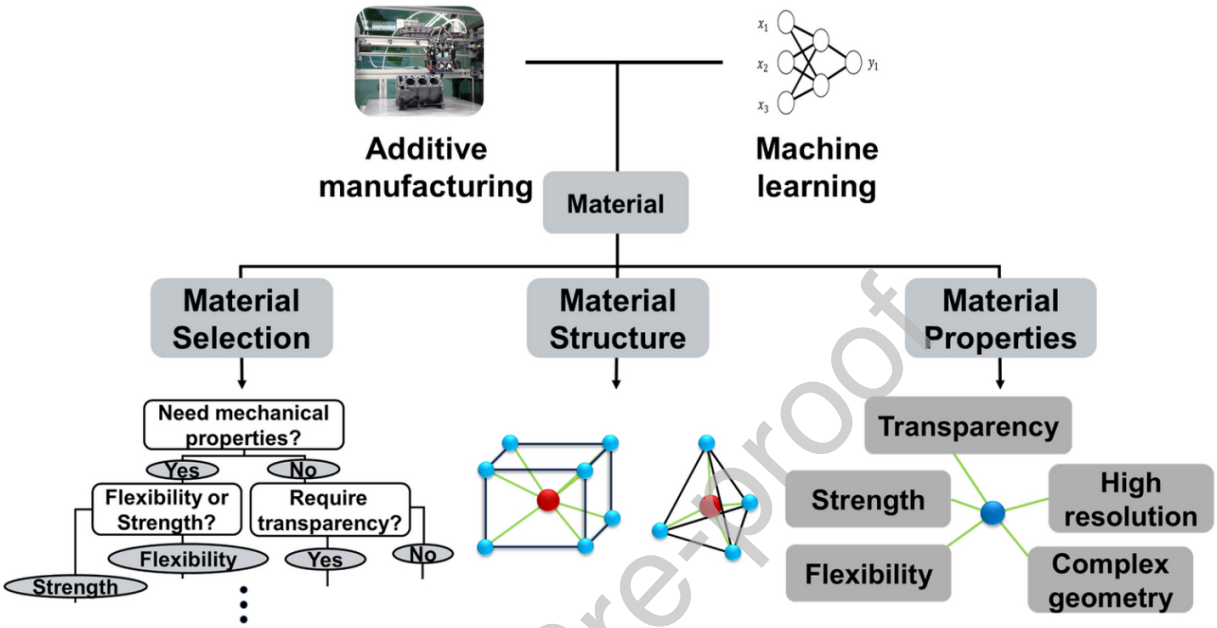

Fig. 7. AI-assisted material development for the AM process [153].

7. Process Monitoring and Control

Process control and monitoring improve when AI and ML are integrated into AM systems. ML models can evaluate real-time data from sensors integrated into AM equipment to anticipate and minimize flaws [154]. This predictive maintenance approach minimizes downtime and ensures higher-quality outputs to enhance productivity in AM processes. AI-based systems can analyze sensor data from AM machines in real time, enabling early detection of defects and process deviations [135, 155]. For example, ML algorithms can analyze data from temperature sensors, laser power, and melt pool dynamics to identify patterns indicative of potential defects [137]. AI can minimize machine downtime and improve overall productivity by implementing predictive maintenance [156]. Moreover, DL techniques have been employed to process vast amounts of data from manufacturing processes, identifying subtle patterns and anomalies that human operators may overlook [157]. Consequently, AM processes have become more reliable and consistent. DL models can be used to track AM processes in real time and to identify problems that traditional methods may neglect [157]. Real-time flaw detection in AM is also possible using AI-driven computer vision systems. AI-powered image recognition can be used to inspect printed parts that are not visible to the naked eye [158]. Moreover, sensors and cameras collect data during printing, which AI algorithms analyze to detect anomalies and deviations from desired parameters [159]. AI can implement closed-loop feedback systems that can adjust process parameters in real time to maintain optimal manufacturing conditions. DL can automate the inspection process, identifying defects that must be addressed in post-processing [160]. AI can also predict the properties of the final product based on process data, helping to plan post-processing steps more effectively [161]. Integrating AI, ML, and DL into AM shows significant promise for

Journal Pre-proof

streamlining processes, enhancing quality, and reducing costs [162]. Manufacturers can improve AM workflow efficiency and creativity by utilizing these technologies. Process monitoring and management of the AM process utilizing AI, ML, and DL improve product quality, lower scrap rates, and minimize production disruptions [143]. Complex picture data can be processed and interpreted using DL approaches, especially convolutional neural networks (CNNs), making it possible to identify flaws and take prompt remedial action [163]. CNNs are widely used for image analysis and have been used to detect defects in AM parts through image-based inspections. They can learn to identify various types of defects from a large set of labeled images, thereby enabling real-time quality control. Consequently, post-processing becomes less essential, and consistent quality is guaranteed [164]. CNNs can analyze images of printed parts to identify surface defects such as cracks, voids, and surface roughness. These techniques provide a nondestructive and automated approach to quality assurance, reducing the reliance on manual inspection [165]. By training DL models on large datasets of annotated images, these systems can achieve high accuracy in defect detection, reducing the need for manual inspection and improving overall quality control [166]. Figure 8 illustrates the use of intelligent classification techniques for powder-bed fusion AM process monitoring [167].

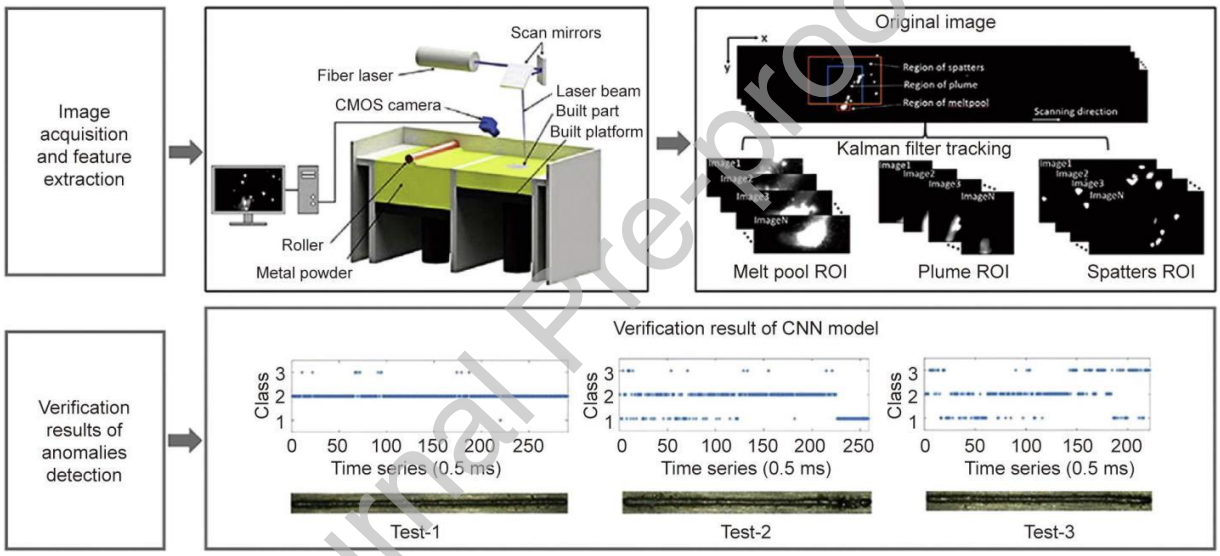

Fig. 8. Powder-bed fusion AM process monitoring using intelligent classification methods [167].

8. Supply Chain and Workflow Optimization

AI can optimize the AM supply chain, ensuring that materials and components are available when needed and reducing inventory costs [168]. In addition, AI-driven systems can automate various aspects of the AM workflow, from design to final product delivery, improving efficiency and reducing human error. Moreover, AI can optimize logistics routes and schedules using machine learning algorithms, thereby reducing transportation costs and delivery times [169]. ML can evaluate and select suppliers based on complex algorithms by considering variables such as cost, reliability, and quality [98]. ML can also be used to create intelligent inventory systems that optimize stock levels using ML algorithms, which reduces overstocks and stockouts. ML algorithms can examine past sales data, industry trends, and outside variables to predict future demand in the supply chain process precisely [170]. DL can also anticipate travel times more accurately and identify supply chain interruptions. DL can evaluate large volumes of supplier data, which aid in making strategic decisions to forecast dangers and possibilities [171]. By

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considering a wide range of factors and their interdependencies, DL can improve the accuracy of inventory projections [172]. Integrating AI, ML, and DL into the AM processes can significantly optimize supply chains and workflows, leading to enhanced efficiency, reduced costs, and improved product quality [173]. Furthermore, this integration can significantly optimize supply chains and workflows, leading to enhanced efficiency, reduced costs, and improved product quality [174]. Companies can remain competitive in the rapidly evolving Industry 4.0 era by employing these innovative technologies. Figure 9 illustrates AI use in the supply chain management of AM systems [175].

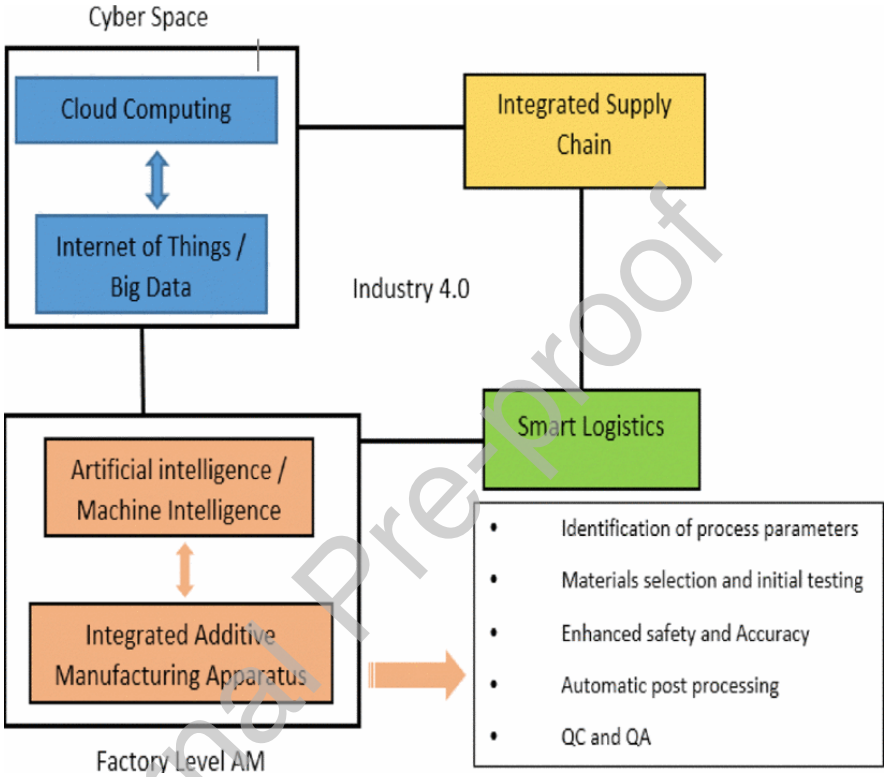

Fig. 9. Application of AI in the supply chain management of AM systems [175].

9. Workflow Automation

AI, ML, and DL are used in AM workflow automation to improve accuracy and efficiency. AI systems make judgments in real time by analyzing large datasets, identifying trends, and optimizing design and production processes. ML models can continuously improve the quality and speed of manufacturing by predicting machine behavior, detecting anomalies, and suggesting corrective actions [176]. DL algorithms that use complex neural networks can significantly improve automation by providing advanced pattern recognition and adaptive learning capabilities. [177]. Integrating AI into AM workflows automates repetitive tasks such as design iterations, process monitoring, and post-processing, freeing up human resources for more complex decision-making and innovation [159]. Moreover, robotic process automation can automate repetitive tasks in the manufacturing process, such as order processing and material handling [67]. AI can also facilitate the seamless integration of AM processes and enterprise resource planning (ERP) systems for better coordination and visibility [178]. Workflow automation in AM processes utilizing AI, ML, and DL has several benefits, including improved operational efficiency, fewer human errors, and faster turnaround times [179]. Furthermore, DL models, particularly convolutional neural networks (CNNs),

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effectively identify defects in printed parts through image analysis [180]. By training on vast datasets of images showing both defective and nondefective parts, these models can detect anomalies such as porosity, cracks, and surface irregularities [181]. Automated defect detection systems can alert operators in real time, enabling immediate corrective actions and reducing the likelihood of defective parts reaching the end user. AI and ML facilitate AM integration into the broader supply chain, as intelligent systems can manage inventory, predict demand, and optimize logistics to ensure that materials and parts are available when needed [182]. Manufacturers can reduce lead times and costs and respond more quickly to market changes by automating these supply chain processes [183]. Autodesk employs generative design algorithms that utilize AI to optimize structures by proposing innovative, lightweight designs that would be challenging to conceive manually [184]. Similarly, General Electric applies ML to predict and correct defects during the printing process, leading to significant improvements in quality and efficiency [185]. Consequently, these technologies collectively streamline the entire workflow, from design and prototyping to production and quality control, leading to reduced costs, minimized errors, and accelerated innovation in AM.

10. Scalability and Flexibility

AI-driven AM systems can easily scale from prototype to production, adapting to changing demands and technological advancements without significant retooling or reprogramming. Scalability and flexibility in AM have been significantly enhanced by integrating AI, ML, and DL [186]. These technologies enable AM systems to adapt and optimize production processes in real time, accommodating a wide range of materials and complex geometries [187]. Large datasets can be analyzed using AI algorithms to forecast and enhance AM process performance. In addition, ML models constantly learn from production data to identify trends and abnormalities that can enhance the performance of decision-making processes [188]. Advanced neural networks in DL approaches enable the autonomous development of design alternatives and item customization to satisfy specific specifications, further refining this process [189]. AI can tailor material properties to specific applications or customer requirements, thereby enabling more personalized manufacturing solutions [19]. Therefore, this integration of AI, ML, and DL into AM improves the scalability of production by allowing for more efficient and precise operations and enhances flexibility by enabling rapid adjustments to changing design and production needs, ensuring that AM can effectively satisfy the demands of various industries [71]. Figure 10 illustrates the AM cycle from the design stage to the finished goods that can be used in the cloud manufacturing environment owing to the connection between cloud computing and IoT. [190].

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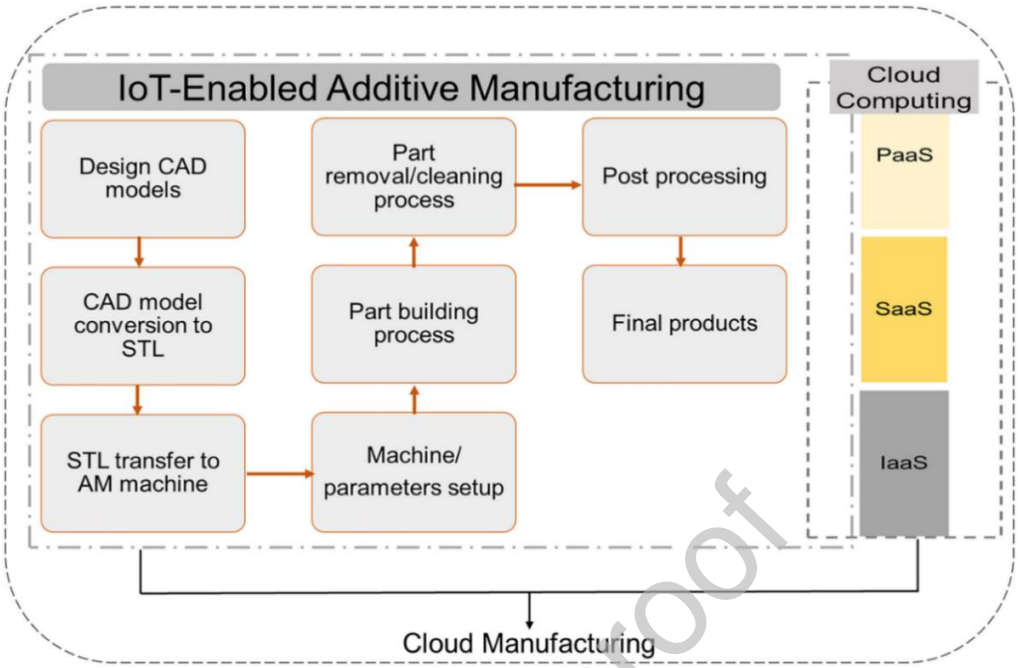

Fig. 10. The AM cycle from design to finish that can be used in the cloud manufacturing environment owing to the connection between cloud computing and IoT [190].

11. Innovation Acceleration

AI, ML, and DL technologies are major drivers of innovation acceleration in AM. AM can accelerate innovation cycles by analyzing and optimizing the materials and processes using AI, ML, and DL [111, 191]. These advancements revolutionize production processes by optimizing material usage, enhancing design complexity, and improving overall efficiency [103]. Moreover, AI algorithms can analyze vast datasets to predict optimal printing parameters, thereby ensuring higher precision and reliability in part production. ML models enable adaptive control systems to continuously learn from operational data, refining manufacturing techniques in real time. DL algorithms can be further improved by automating defect detection and quality assurance, thereby minimizing errors and reducing production costs [105]. These technologies work together to create a new era of flexible, sustainable, and financially feasible production solutions through AM [192]. These technologies accelerate the adoption of new materials and processes, enable rapid prototyping, and iteratively enhance designs. AI, ML, and DL integration in AM offers several advantages, including improved inventory management, lower holding costs, and better production alignment with market demand [193]. AI-powered simulations can predict the performance of new designs before they are physically printed, thereby enabling faster experimentation and optimization. Consequently, manufacturers can introduce novel products quicker, which reduces the time and expense involved in traditional trial-and-error procedures [194]. Furthermore, AI and ML can facilitate the customization and personalization of products manufactured through additive processes [159]. AI algorithms can generate unique designs customized to the requirements of each client by examining consumer data and preferences [195]. This feature is especially useful in sectors such as healthcare, where AM technology can be used to design and produce prostheses and implants customized for each patient. Having the capacity to provide personalized solutions quickly improves client satisfaction and creates new business prospects. ML models can also forecast the performance

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and failure points of prototype designs under various circumstances before they are manufactured, enabling incremental improvements [189]. This reduces the time and expense incurred by traditional trial-and-error techniques in manufacturing procedures.

12. Sustainability in AM

AI, ML, and DL are revolutionizing AM owing to their sustainability potential. These technologies optimize design processes, reduce material waste through predictive modeling, and enhance energy efficiency in production [196]. AI-driven simulations enable rapid prototyping and iterations, leading to lighter and more durable components that require fewer resources [34]. ML algorithms analyze vast datasets to improve material selection and usage, minimizing the environmental impact while maximizing performance [147]. In addition, DL techniques refine AM processes by identifying flaws early, ensuring higher quality outputs and less scrap [197]. Using AI to minimize waste and energy consumption in AM contributes to greener manufacturing practices, as AI can optimize material usage, thereby reducing waste and improving sustainability [198]. ML models can also optimize the recycling process of used materials, ensuring high-quality outputs [199]. Consequently, AI, ML, and DL provide advanced sustainable offerings in AM, decreasing environmental pollution associated with AM processes. Figure 11 describes sustainability in AM [200].

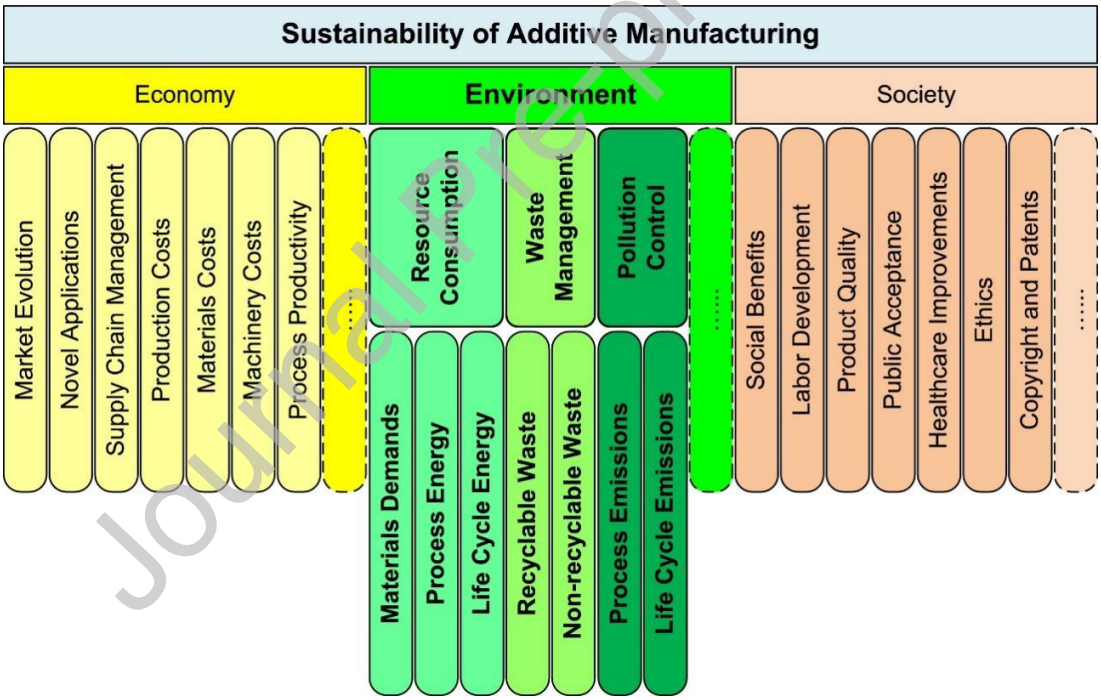

Fig. 11. Sustainability in AM [200].

13. Adaptive Learning

Adaptive learning is a dynamic process in which models or systems change their behavior in response to fresh information. Integrating AI, ML, and DL into AM has significantly improved the flexibility and efficiency of manufacturing processes [118]. AI algorithms analyze vast amounts of data from sensors embedded in 3D printers, adjusting parameters in real time to enhance print quality and efficiency [96]. ML models can predict potential defects or material inconsistencies before they occur, thereby

Journal Pre-proof

ensuring high precision and reducing material waste [201]. DL techniques enable AM systems to learn from experience, continually improving their performance based on feedback loops from production outcomes [162]. This synergy empowers AM to evolve beyond prototyping into mainstream production, offering customizable, cost-effective solutions across industries from aerospace to healthcare [202], developing ML models that can adapt to new materials and designs without extensive retraining. ML models can analyze historical data to predict future trends and identify potential bottlenecks in manufacturing processes. AI can simulate various production scenarios to assist managers in making informed decisions [191]. Therefore, better strategic planning, risk mitigation, and enhanced decision-making can be implemented by modifying the AM process using AI, ML, and DL [2, 71]. Integrating AI, ML, and DL into AM offers further innovation as these technologies develop, accelerating manufacturing to unprecedented levels of efficiency and flexibility [203]. Slicing in AM refers to the process of dividing a 3D model into thin layers for sequential printing. Adaptive learning enhances this process by dynamically adjusting the slicing strategy based on the complexity of the geometry and material properties [204]. Similarly, adaptive toolpath generation algorithms optimize the path followed by the printing nozzle to minimize printing time and material usage while maintaining structural integrity. These improvements result in more efficient and cost-effective manufacturing processes [143]. Adaptive learning facilitates personalized manufacturing in which products can be tailored to individual specifications [205]. By analyzing customer data and preferences, AI-driven systems can adapt their design and manufacturing processes to produce customized parts [206]. This is particularly valuable in industries such as healthcare where patient-specific implants or prosthetics can be created [207]. Adaptive learning models continuously refine the understanding of individual needs, leading to more precise and effective personalized solutions.

14. Conclusion

AM has been modified owing to the capabilities of AI, ML, and DL, providing unprecedented levels of efficiency, inventiveness, and precision. These technologies have significantly improved the entire AM process, from material selection and design optimization to production and post-processing. Significant prospects for improving material creation and streamlining production processes have emerged from integrating AI, ML, and DL into AM. These technologies can help create more dependable and efficient AM systems by improving process optimization, quality control, predictive maintenance, and material development. Real-time process monitoring and quality control are essential to ensuring the reliability and consistency of AM products. AI and ML algorithms can be used to examine and analyze data from sensors during AM processes to identify anomalies and variations during printing. AI algorithms can optimize manufacturing designs, leading to material savings and enhancing structural integrity. AI systems can produce ideal design solutions given a set of parameters and specifications. AI can forecast when equipment will require repair, saving downtime and increasing productivity. Real-time flaw detection in manufacturing can be achieved with AI-driven computer vision systems. By considering several variables, including material supply, order priority, and equipment availability, ML models can improve production schedules. AI can enable the mass customization of products by efficiently managing the complexity of customized orders. However, high-quality data are essential for training accurate AI and ML models. Collecting and curating such data can be challenging. Integrating AI/ML solutions into existing AM workflows requires significant investments and expertise. Ensuring that AI/ML models can scale to different types of AM machines and processes is critical to their widespread adoption. The development of new materials is a key area in which AI, ML, and DL have made significant contributions. These technologies can analyze vast datasets to

Journal Pre-proof

identify the relationships between material compositions and their properties. ML algorithms can then suggest new material formulations that satisfy specific performance criteria. ML models can also predict the performance and failure points of these designs under various conditions, enabling iterative improvements before the physical prototypes are built. This reduces the time and cost associated with the trial-and-error methods in traditional manufacturing processes.

AI algorithms analyze vast datasets to predict material behavior and optimize designs to ensure that the components produced are lighter, stronger, and more cost-effective. ML models improve manufacturing consistency by learning from real-time sensor data, thereby reducing defects and waste. DL techniques such as generative design create intricate geometries that were previously impossible, pushing the boundaries of what can be produced with AM. Together, these advancements are ushering in a new era of customizable, sustainable manufacturing, in which complex parts can be produced on-demand with minimal human intervention. As AI, ML, and DL continue to evolve, their integration into AM promises to redefine industries, accelerate product development cycles, and unlock previously unimaginable design and production possibilities. Despite the promising advancements, AI, ML, and DL integration into AM faces several challenges. These include the requirement for sizable and superior-quality datasets, the intricacy of AM procedures, and the computational power needed to train advanced models. Subsequent investigations should concentrate on formulating resilient and effective algorithms, enhancing techniques for data gathering, and investigating the amalgamation of these technologies with additional Industry 4.0 endeavors. Manufacturers can achieve higher productivity, improved quality, and more creative solutions in the AM space by utilizing this technology. AI, ML, and DL integration into AM has enormous potential for process optimization, quality improvement, and cost reduction. AI, ML, and DL integration in AM has several benefits, including enhanced customer satisfaction, increased sales, and a competitive advantage.

It is imperative that the necessary standards and fault-detection procedures be addressed when integrating AI techniques, such as ML and DL, into AM. AI can optimize the AM process by predicting potential defects and enhancing quality control. However, without standardized protocols, integration can lead to inconsistent results. Therefore, establishing clear standards for AI applications in AM is vital to ensure reliability, reproducibility, and safety in manufacturing. In addition, AI-driven defect detection should be rigorously tested and validated to minimize errors and ensure that the final products satisfy the desired specifications and quality standards.

15. Future Research Work Directions

Integrating AI, ML, and DL into AM processes and supply chains could significantly increase productivity, reduce costs, and improve product quality. These technologies will likely have a greater influence on AM as they develop, resulting in more creative and efficient production techniques. Obtaining a thorough understanding of the thermal-cycle heat interaction behavior during the AM process is crucial before establishing data-driven AI models in future research. The complex relationship between thermal dynamics, material design, and the resulting properties should be fully elucidated to enhance the reliability and accuracy of AI predictions. In addition, if AI techniques can be effectively applied to the real-world conditions of AM processes, developing corresponding standards and addressing potential defects are necessary. This ensures that AI integration in AM optimizes manufacturing outcomes but maintains the quality and integrity of the final products. Future research directions can also include developing more efficient algorithms that require less data, integrating multi-physics simulations to enhance model accuracy, and creating user-friendly software tools that make these technologies accessible to a broader range of designers and engineers. The ultimate goal of integrating AI, ML, and DL into AM is to develop fully autonomous AM

Journal Pre-proof

systems capable of self-optimization, self-monitoring, and self-repair. Autonomous AM systems can leverage AI algorithms to continuously learn and adapt to changing process conditions, thereby improving efficiency, quality, and productivity. Achieving full autonomy in AM requires advancements in AI algorithms, sensor technologies, and real-time control systems. Combining AI, ML, and DL with other advanced technologies such as physics-based modeling and simulation can lead to hybrid approaches that leverage the strengths of different methods. As the AM landscape continues to evolve, several key areas offer promising avenues for future research, including the following:

- Integration with IoT: Combining AI/ML with IoT for smarter, interconnected manufacturing systems.
- Advanced Simulation: Using AI to enhance simulation models, thereby providing better predictions of material behavior and process outcomes.
- Collaboration Platforms: Developing collaborative AI platforms where multiple stakeholders can contribute to material development and process optimization.
- Developing advanced algorithms for real-time optimization of printing parameters to improve quality and reduce material waste.
- Utilizing generative design techniques powered by AI to create more efficient and innovative structures.
- Implementing predictive maintenance models to foresee equipment failures and minimize downtime.
- Applying ML algorithms to predict and tailor material properties, ensuring that the final products meet specific performance criteria.
- Investigating AI use in developing new composite materials that offer superior mechanical and thermal properties.
- Utilizing DL models to analyze the microstructure of printed materials and optimize their composition for better performance.
- Developing advanced image recognition algorithms for real-time defect detection during printing.
- Implementing AI-based systems for nondestructive testing and evaluation of printed parts.
- Utilizing DL techniques to create predictive models that can identify potential defects based on historical data and real-time monitoring.
- Exploring the potential of AI to design and manufacture bespoke medical implants and prosthetics tailored to individual patients.
- Investigating AI use in the production of customized consumer goods, such as fashion items and household products, based on individual preferences and needs.
- Developing AI-driven platforms that allow end users to participate in the design process, ensuring their specific requirements are met.
- Researching AI techniques to optimize material usage and minimize waste, thereby contributing to more sustainable manufacturing practices.
- Developing ML models to assess and reduce the carbon footprint of AM processes.
- Exploring AI use in recycling and reusing materials in AM, thereby promoting a circular economy approach.
- Investigating the potential of AI and ML in creating fully autonomous AM systems that can self-monitor, self-correct, and self-optimize.

Journal Pre-proof

- Exploring AI use in creating digital twins in AM processes and providing a virtual replica for real-time monitoring and optimization.
- Developing ML algorithms to facilitate seamless communication and data exchange between AM machines and other Industry 4.0 components.
- Developing AI systems capable of analyzing vast amounts of data generated during the AM process to extract valuable insights and improve decision-making.
- Implementing ML models to identify patterns and correlations in AM data, thus leading to better understanding and control of the manufacturing process.
- Exploring DL technique use for advanced data analytics, enabling predictive modeling and intelligent decision support systems.

In conclusion, the fusion of AI, ML, and DL with AM has immense potential to revolutionize the field. By addressing these future research directions, the AM industry can achieve unprecedented levels of efficiency, customization, and sustainability, paving the way for a new era of manufacturing innovation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

CRediT authorship contribution statement

Mohsen Soori: Conceptualization, Methodology, Investigation, Data Curation, Writing Original draft, Visualization.

Fooad Karimi Ghaleh Jough: Resources, Formal analysis, Review & editing.

Roza Dastres: Conceptualization, Review & Editing.

Behrooz Arezoo: Supervision, Review & Editing.

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I confirm that there is no conflict of interest regarding the submitted manuscript with title of **'Additive manufacturing modification by artificial intelligent, machine learning and deep learning, A Review' to the Additive Manufacturing Frontiers.**